

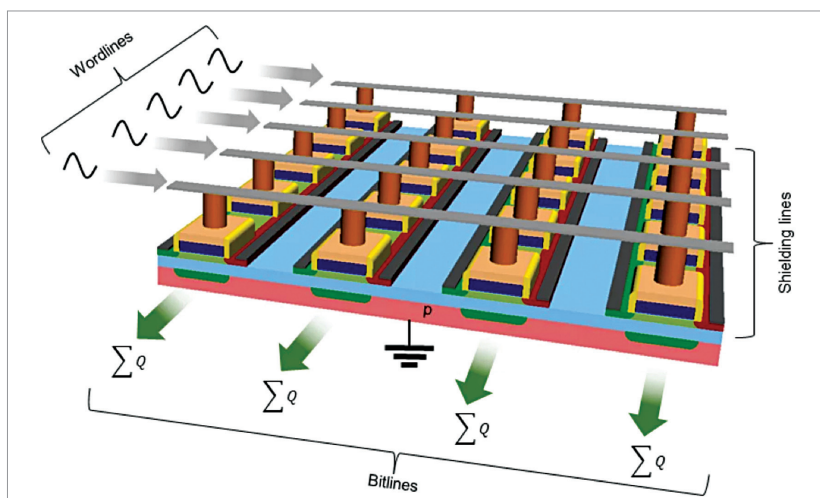
Personalised exosuit for activity-specific assistance

People walk at inconsistent speeds, which changes with respect to various external conditions. Additionally, walking is influenced by muscle strength, and sometimes by neural or muscular disorders such as stroke or Parkinson's disease. Designing custom wearable robotics to assist or augment walking in real-world conditions is a major challenge. Now, researchers from the Harvard John A. Paulson School of Engineering and Applied Sciences (SEAS) and Harvard's Wyss Institute for Biologically Inspired Engineering have developed a new approach in which robotic exosuit assistance can be calibrated to an individual and adapted to a variety of real-world walking tasks in a matter of seconds. The bio-inspired system uses ultrasound measurements of muscle dynamics to develop a personalised and activity-specific assistance profile for users of the exosuit.



Researchers developed a new approach in which robotic exosuit assistance can be calibrated to an individual and adapted to a variety of real-world walking tasks (Credit: <https://wyss.harvard.edu>)

Ultra-low-power capacitive memory technology



Proposed crossbar structure for highly efficient vector matrix multiplications (Credit: www.mpi-halle.mpg.de)

Much effort is being made to develop new types of memory devices that can revolutionise the computing paradigm of the past half century. Major breakthroughs are required, particularly in deep learning as a subset of artificial intelligence. Recently, a team of scientists at the Max Planck Institute of Microstructure Physics, Halle (Saale), and SEMRON GmbH, Dresden, Germany, have successfully demonstrated a unique capacitive memory technology with unprecedented energy-efficiency. The technology can easily be implemented using conventional silicon manufacturing processes and systems, thereby paving the way to its rapid commercialisation and adoption in many application domains.

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